

Kernel PLS for binary classification

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1 Introduction

The problem in this assignment is to train a K-PLS model to predict a class label for an unknown data point. The problem in the ground level is a binary classification problem, but since we are dealing with highly collinear data and noisy data, we must use some more advanced methods.

Ordinary PLS decomposes the feature space X as

$$X = TP^\top + E \quad (1)$$

where T are the latent scores, P the variable loadings and E the model residual. In K-PLS we work in kernel space where

$$K = \Phi(X)\Phi(X)^\top \quad (2)$$

defines the kernel matrix and $\Phi(\cdot)$ is the mapping operator. The kernel matrix replaces the inner product in PLS with $K = XX^\top$. Latent variables are extracted from K to maximize covariance with the response y and can then be used for regression or classification.

Regression in kernel space is performed by stacking the latent scores $T = [t_1, \dots, t_A]$ and computing

$$B = T(T^\top KT)^{-1}T^\top Y. \quad (3)$$

For a new sample x^* , the centered kernel vector

$$K^* = [k(x^*, x_1), \dots, k(x^*, x_n)]^\top \quad (4)$$

is formed, and the prediction for an unknown sample is

$$\hat{y}^* = K^* B. \quad (5)$$

2 Data

Our data set contains of 87 instances of diagnostic parameters for a liquid ultrasonic flow meter, having 37 attributes and two classes. The class values describe flow meter states, which are either healthy or under installation effects.

The data set is normalized and split into training and testing sets using 75/25 split. The

correlation between variables are investigated with a correlation matrix, which is presented in Figure 1. From the correlation matrix we can see that there are strong correlations among multiple features. This indicates that PLS is more than required for this issue, since data is highly collinear. We will use K-PLS for the solution, since the relation between features and the predicted value (binary class value) is nonlinear.

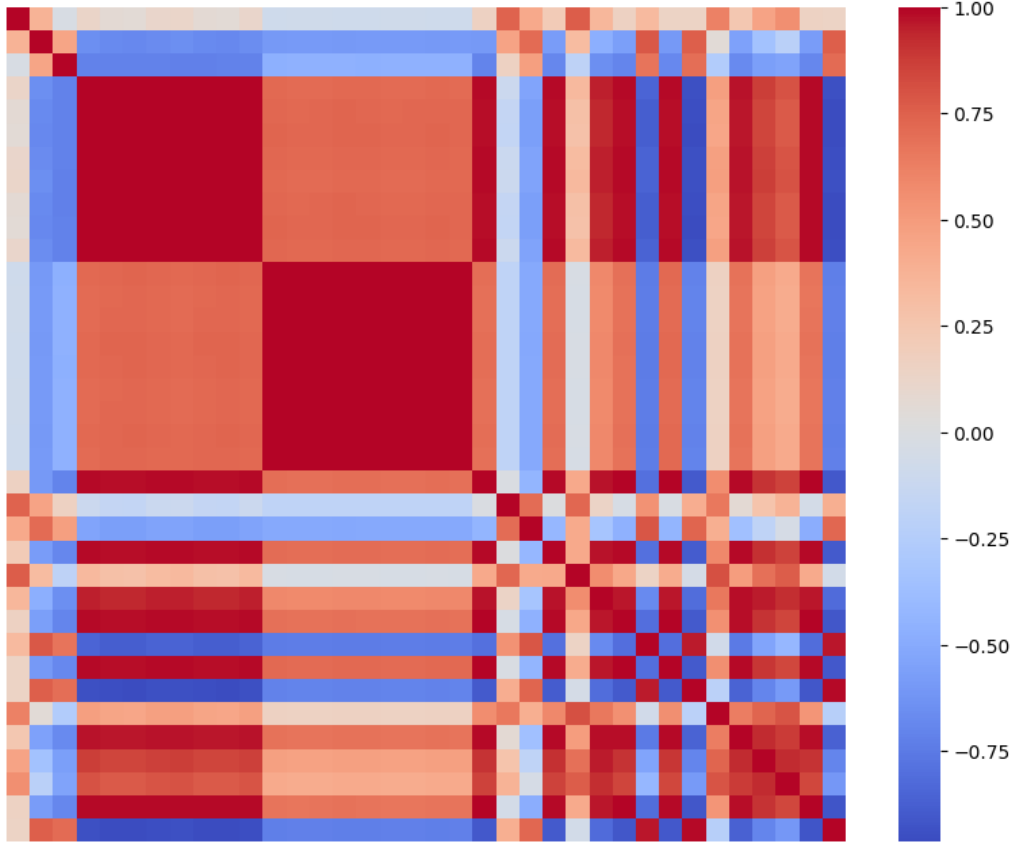


Figure 1: Correlation matrix for variables in the liquid ultrasonic flow meter dataset. High positive correlations are presented with dark red, high negative correlations with dark blue and low correlations with gray.

3 Results

Data was loaded in Python and `scikit-learn` library was used for computing K-PLS. Gaussian kernels were used for the K-PLS with $\gamma = 0.1$. The number of components used for K-PLS was determined using cross-validation. In cross-validation, different number of components were used for ordinary PLS regression, and the RMSE values were computed for each number of components. The cross-validation results can be seen in Figure 2. Based on the cross-validation results, 29 components were used for K-PLS.

K-PLS regression model was fit for the training data, and the validation was done with the test data. The accuracy for training accuracy was 100% (perfect separation), while the test accuracy was 86%. When looking at the class-wise accuracies, the model predicted perfectly healthy data samples (100% accuracy), whereas some of the samples with installation effects

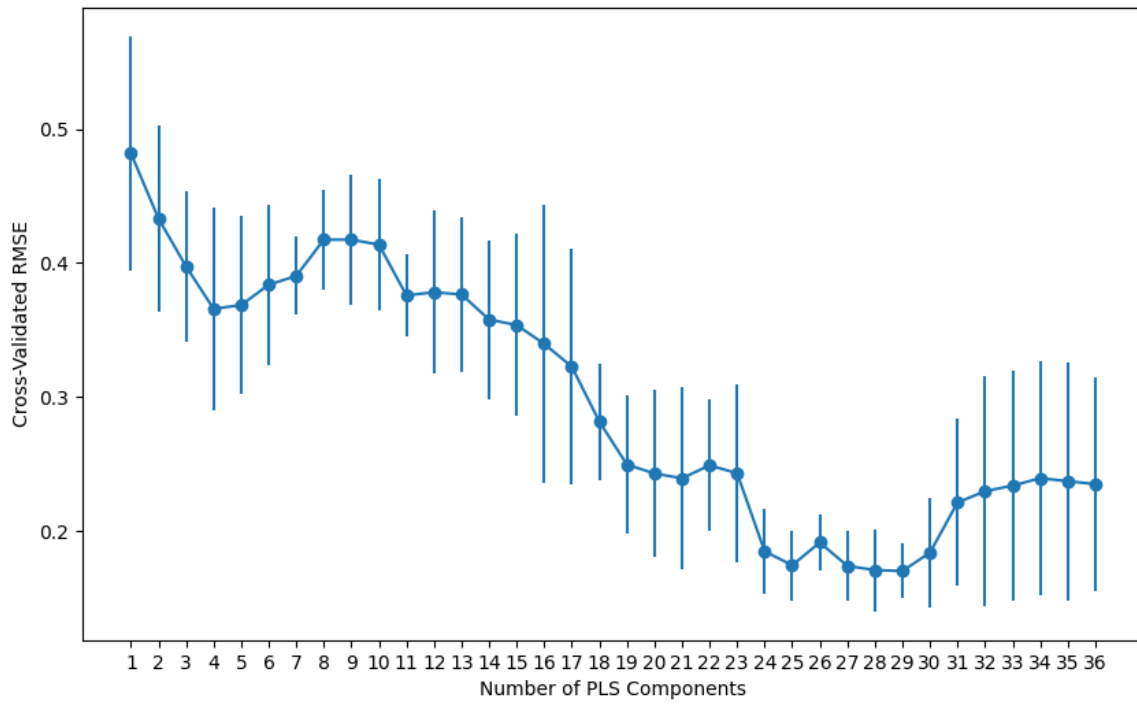


Figure 2: Cross-validation results for the normalized data. The lowest RMSE value is achieved when using 29 components.

were misclassified as healthy samples (83% accuracy). The class-wise results are shown as a confusion matrix in Figure 3.

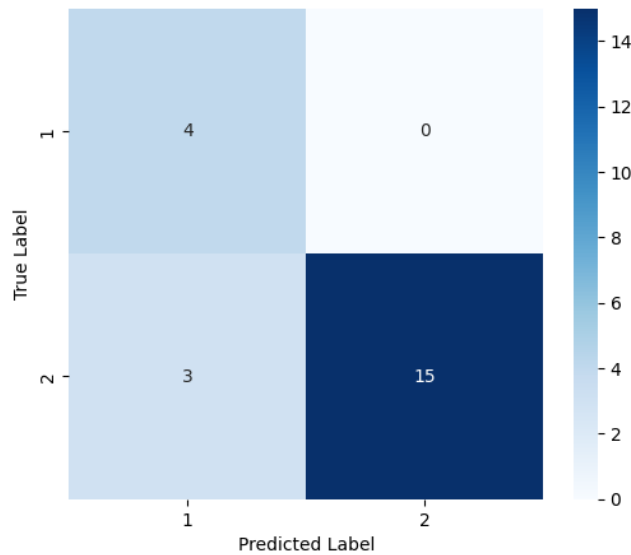


Figure 3: Confusion matrix on the prediction results.

4 Conclusions

With 86% total accuracy the model predicts the state rather well. However, because the data size is so small, the division to the train and test sets play a big role in classification. Better understanding about the goodness of the K-PLS results could be achieved by having a baseline model as a comparison, for example a simple neural network or random forest.

Class imbalances in the test set also may have affected the results. There are only 4 healthy samples in the test set, and 18 samples with installation effects. By having a class balance in both train and test sets, more robust results can be achieved. If the classes are heavily unbalanced, the model might learn to predict unknown samples as the more frequent class.

To summarize, the model works with decent accuracy for the given data set. However, more data is needed in order to determine the true goodness of the model.